

59th IAA SYMPOSIUM ON SAFETY, QUALITY AND KNOWLEDGE MANAGEMENT IN SPACE ACTIVITIES (D5)

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Author: Mr. Saatvik Sumanta
India

Mr. Kartikey Prasad
India

Ms. Anshika Ranjan
India

Mr. Austin George Francis
India

PROBABILISTIC SOLAR ERUPTION FORECASTING FOR RADIATION RISK MITIGATION IN DEEP SPACE MISSIONS

Abstract

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Human exploration beyond low Earth orbit is accelerating, with sustained lunar missions under NASA's Artemis program and planned crewed expeditions to Mars becoming realistic objectives. As missions operate outside Earth's magnetospheric protection, radiation driven by solar flares and coronal mass ejections (CMEs) becomes a critical safety concern. Solar energetic particle (SEP) events can endanger astronauts, degrade spacecraft systems, and disrupt mission timelines. Improving forecast reliability, lead time, and uncertainty quantification of solar activity is therefore essential for robust risk management in deep-space operations.

NASA's Solar Dynamics Observatory Machine Learning dataset (SDOML v2) provides over ten years (2010-2020) of aligned, degradation corrected, full disk solar observations, including imaging channels from the Atmospheric Imaging Assembly spanning extreme ultraviolet, ultraviolet, and visible wavelengths, along with vector magnetogram components (B_x , B_y , B_z) from the Helioseismic and Magnetic Imager. These multi-modal observations capture the coupled radiative and magnetic evolution of active regions across Solar Cycle-24 and the rising phase of Solar Cycle-25, enabling spatio-temporal modeling of solar dynamics.

This paper proposes a probabilistic spatio-temporal framework that ingests 24-48 hour multi-channel sequences and produces forecasts of active-region evolution and eruptive likelihood over 6-48 hour horizons. The architecture employs a video-transformer or diffusion-based backbone with physics-informed constraints, including temporal magnetic flux continuity and radiative-intensity consistency. Instead of deterministic outputs, the model generates ensemble predictions with calibrated epistemic and aleatoric spatial uncertainty maps. These are used to estimate flare probabilities and CME likelihood indicators inferred from disk-based eruption proxies such as filament destabilization and coronal dimming.

The framework is evaluated across multiple phases of Solar Cycle-24 and 25 using structural similarity metrics, event-based precision-recall analysis, and comparison with operational NOAA forecasts. By providing uncertainty calibrated forecasts, the system supports radiation risk assessment workflows and strengthens the reliability of software-driven tools for safety-critical deep-space missions.